

Maths

Probability

1. Probabilistic graphical models can only do 2 things:
 1. Inference: calculating a probability of some data modelled as RV using the model $\mathcal{M}$:
 1. evaluation $P(obs|state)$,
 2. decoding $P(state|obs)$
 2. Learning model parameters: refers to finding the graphical model $\mathcal{M}$, given some data $\mathcal{D}$.
 1. MAP (maximum a posteriori)
 2. Maximum likelihood
 3. expectation maximization
 4. forward-backward algorithm
2. Conditional probability can be computed using:
 1. Bayes rule
 2. Enumeration
3. θ is model parameter.
4. For example RV x is binomial, x is the number of times a coin landed heads in N trials.

1.
$$P(x = X|\theta = p) = \binom{N}{x} p^x (1-p)^{N-x} \quad (\text{likelihood})$$

5. if RV x is single variable Gaussian, for modeling a real value, e.g. temperature sensor reading.

1.
$$P(x = X|\theta = (\mu, \sigma^2)) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (\text{likelihood})$$

6. Note: Notice that in both cases the RV x ended up in the exponent.

7.
$$p(\theta|x = D) = \frac{p(x|\theta)p(\theta)}{p(x)}$$

$p(x|\theta)$ is the likelihood, $p(\theta)$ is the prior distribution, $p(\theta|x = D)$ is the posterior, and $p(x)$ is the normalization term.

Probabilistic Inference Methods

1. Maximum Likelihood Estimation (MLE)
 1. $x^* = \arg \max p(d|x)$.
 2. Likelihood: $p(d|x)$
 3. x^* is the most probable value of x .
 4. $\{x\}$ is a set of possible values of x (the candidates of x).
 5. $\arg \max$ to return the value of x that maximizes $p(d|x)$.
 6. The weakness of MLE is that it allows impossible candidates, as it treats candidates uniformly.
 7. The intuitive meaning of MLE:
 1. Take each possible illness e.g. $x=\text{flu}$.
 2. Verify if it fits to the symptoms $\rightarrow$ probability of x .
 3. Decide the illness based on the highest probability.

2. Maximum A Posteriori (MAP)

1.
$$\begin{aligned} x^* &= \arg \max p(x|d) \\ &= \arg \max p(d|x)p(x) \end{aligned}$$

. Likelihood:

2. MAP treats the candidates differently: $p(x = \text{flu}) > p(x = \text{cancer})$
3. Main criticism of MAP is how to determine $p(x)$.

3. Full Bayesian Inference

1. $p(x|d) = \frac{p(d|x)p(x)}{p(d)}$. $P(d)$ is intractable to solve.
2. This is the ideal and most robust inference that besides providing a prediction on the value of x , it also provides the uncertainty, or how high/low the probability of the prediction.

3. It's possible because of $p(d)$, the evidence. This will normalize the product of $p(d|x)p(x)$.
4. However to obtain $p(d)$ is analytically intractable, and in discrete, it can be prohibitively expensive.
4. Example: According to our observation, a patient has a set of symptoms: $d : \{\text{stiff neck, coughing, headache, fever}\}$. We also have a set of possible illness: flu, meningitis, infection and cancer.

Lagrange Multiplier [3]

Hungarian Matching [2]

References:

1. Harrison H. Barrett, Kyle J. Myers. Foundations of Image Science.
2. Discrete Mathematics, John Dossey, 1987.
3. Machine Learning Fundamentals: A Concise Introduction, Hui Jiang.
4. [Neural Message Passing](#)
5. Lecture 7, Stereo + MRF, [EE5731 Visual Computing](#)
6. Intro Prob and Statistics, Paul L. Meyer.
- 7.